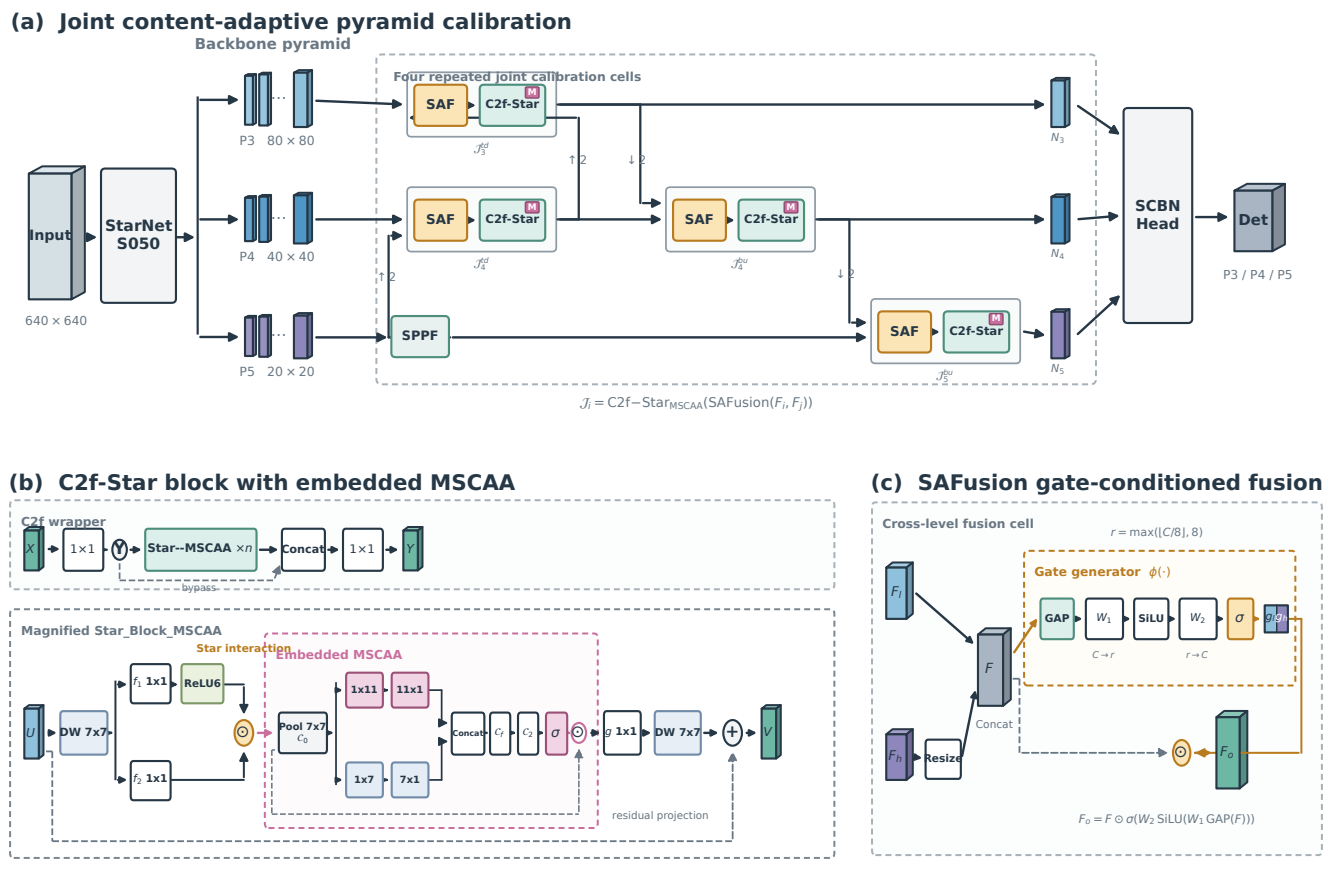
(a) Joint content-adaptive pyramid calibration
Backbone pyramid
Four repeated joint calibration cells
Input
640 × 640
StarNet S050
P3 80 × 80
P4 40 × 40
P5 20 × 20
SAF
C2f-Star
M
SPPF
↑ 2
↓ 2
SCBN Head
Det
P3 / P4 / P5
$\mathcal{J}_i = \mathrm{C2f{-}Star}_{\mathrm{MSCAA}}(\mathrm{SAFusion}(F_i, F_j))$
(b) C2f-Star block with embedded MSCAA
C2f wrapper
1×1
Star--MSCAA ×n
Concat
bypass
Magnified Star_Block_MSCAA
Star interaction
Embedded MSCAA
DW 7x7
ReLU6
Pool 7x7
1x11
11x1
1x7
7x1
residual projection
(c) SAFusion gate-conditioned fusion
Cross-level fusion cell
$r = \max(\lfloor C/8 \rfloor, 8)$
Gate generator $\phi(\cdot)$
GAP
SiLU
Concat
Resize
$F_o = F \odot \sigma(W_2\,\mathrm{SiLU}(W_1\,\mathrm{GAP}(F)))$